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ORIGINAL ARTICLE

Effects of peer network intervention through peer-led play on Basic Social Communication Skills of children with Autism Spectrum Disorder in Inclusive Classroom

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Abstract

This study evaluated whether a play-based, teacher-facilitated peer network intervention could improve basic social communication skills of children with autism spectrum disorder (ASD). The study was conducted in a regular classroom, using the principles of peer mediation and an integrated playgroup model. A single-subject multiple baseline design across three participants with ASD and Tau-U were used to determine the size of the effect.The results indicated that basic social communication skills of all three target participants significantly improved from the baseline condition in all three aspects measured: the duration of social interaction, social initiations, and responses to their peers.In the maintenance condition, three target participants retained increased social communication skills yet at lower levels than during the intervention condition. Overall, the results suggested that the intervention package was effective at improving the social abilities of children with ASD and applicable in classrooms. Practices, implications, and recommendations for future research are discussed.

Keywords Peer-mediated · Social interactions · Autism · Communication impairments · Inclusive education

Introduction

A deficit in social communication skills characterizes many individuals with autism spectrum disorder (ASD), which can cause social integration problems and a failure

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to engage meaningfully with peers (American Psychiatric Association, 2013). While children with ASD may experience a range of difficulties related to social integration, challenges around social communication and interaction are common, especially in the school environment (Banda & Hart, 2010). This can lead to detachment, isolation, and further exacerbation of isolating behaviors (American Psychiatric Association, 2013). To improve the social skills and behavior of children with ASD, researchers and practitioners have developed evidence-based practices, revealing a variety of approaches that enhance social skills and communication competence in natural settings (Goldstein, 2002; Kasari & Lawton, 2010; Koegel et al., 2012a, b; Morrison et al., 2001; Reichow & Volkmar, 2010; Wong et al., 2015). This study evaluated the effects of peer network interventions (PNIs) using an integrated peer-led play intervention to assist children with ASD develop social and communication skills.

Peer-mediated interventions

Peer-mediated interventions (PMIs) represent a set of interventions which have been demonstrated as effective procedures for improving the social skills of children with ASD, as well as fostering their involvement in inclusive settings through applying an array of models, techniques, and strategies (Harrell et al., 1997; Watkins et al., 2015). PMIs consist of having typically developing peers who are selected, trained and supervised to teach or support their peers with ASD (Chung et al., 2007; Schmidt & Stichter 2012; Płatos & Wojaczek, 2018). These peers usually play a supportive role, where they try to communicate with their peers with ASD by partnering with them during play and providing support in specific activities, such as completing an assignment (Charlop et al., 2018; Hott et al., 2014). Previous research has suggested numerous positive effects of PMIs on developing different skills in children with ASD, such as communication, play, and social skills Bene et al., 2014; Gunning et al., 2018; Shivers & Plavnick, 2015; Sperry et al. 2010; Watkins et al., 2015). During PMIs, children with ASD practice different skills with their peer-partners, which leads to the development of social skills during the intervention (Watkins et al., 2015) and possible skill generalization (Wang et al., 2011; Watkins et al., 2015). Typically developing peers also benefit from PMI, through increased positive interactions with their peers with ASD and enhanced sensitivity to others (Carter et al., 2010; Gunning et al., 2018).

PMIs first emerged as an intervention for young children with ASD but has since been adapted to children of all ages as well as young adults. Given the well-documented need for early interventions with children with ASD, studies highlighting the effectiveness of PMIs for improving the communication and social skills of preschool children with ASD dominate the literature (Betz et al., 2008; English et al., 1997; Garfinkle & Schwartz, 2002; Katz & Girolametto, 2013; Kohler et al., 2007; Oppenheim-Leaf et al., 2012; Trembath et al., 2009). Other researchers have also reported significant positive effects of PMIs on the social behavior of primary school children with ASD (Banda & Hart, 2010; Harrell et al., 1997; Laushey & Heflin, 2000; Oppenheim-Leaf et al., 2012; Owen-DeSchryver et al., 2008; Thiemann & Goldstein, 2004). More recently, studies have revealed the effectiveness of PMIs for

adolescents Koegel et al., 2012a, b; Stichter et al., 2010; Watkins et al., 2015; Thomas & Bambara, 2020).

PMIs utilize a variety of different techniques in different settings to discern which techniques are most effective at improving the communication and social behavior of different groups of children with ASD. While classrooms serve as the typical PMI setting, researchers have increasingly experimented with other settings such as during lunchtime (Owen-DeSchryver et al., 2008), recess (Harper et al., 2008; Owen-DeSchryver et al., 2008) and play sessions (Katz & Girolametto, 2013; Trembath et al., 2009). Some studies have experimented with the use of siblings as peers to improve play skills among children with ASD (Oppenheim-Leaf et al., 2012) while others have tested the effectiveness of PMIs for teaching children with ASD to use augmented and assistive communication devices (Trembath et al., 2009; Trottier et al., 2011). Researchers have tested the efficacy of proximity-based interventions that bring children with ASD and their peers together, such as joint activity schedules or groups structured around perseverative interests, like lunch clubs (Koegel et al., 2012a, b).

Several systematic reviews in the previous literature have examined the effectiveness of PMIs with ASD. PMIs were found to be highly effective at improving social skills and social interactions among children with ASD (Chan et al., 2009; Zhang & Wheeler, 2011; Watkins et al., 2015; Wang et al., 2011; Gunning et al., 2018) reported that PMIs positively impacted social, communication and play skills of preschool children with ASD in inclusive education. Further, PMIs were identified as effective interventions for enhancing social interaction between children with ASD and their peers in inclusive education (Martinez et al., 2019; Watkins et al., 2019; Aldabas, 2020) found that PMIs improved social behavior and communication skills in young children with ASD. This previous literature indicates that PMIs could be suitable for supporting social development in children with ASD.

One effective intervention that naturally targets social interactions is peer mediation through peer networks. Peer networks consist of small groups of peers who are recruited to facilitate activity engagement, reinforce play behavior, or tutor academics (Bauminger et al., 2008; Harrell et al., 1997; Haring & Breen, 1992; Kamps et al., 1997; Kasari et al., 2006; Parker & Kamps, 2011; Thiemann & Goldstein, 2004). In contrast to peer tutoring, which typically involves pairs of children and emphasizes academic over social support, peer networks are small social groups of typically developing peers and children with ASD (Hott et al., 2014; Kamps et al., 1999). The peer group forms around the children with ASD to promote social and communication outcomes and/or academic support within the classroom throughout the school day (Harrell et al., 1997; Owen-DeSchryver et al., 2008; Carter et al., 2010).

Peers could be selected to provide support for children with ASD to encourage greater integration in social environments (Kamps et al., 2015). Together, these peers participate in training to learn how to conduct PMIs with children with ASD in a naturalistic setting. PNIs typically combine adult facilitation, repeated social learning opportunities in natural settings with peers, and active peer mediation with children with ASD (Kamps et al., 2015). As reported in previous studies (Kamps et al., 2015; Kasari et al., 2011; Mason et al., 2014; Thiemann & Goldstein, 2004), PNIs, which are a form of PMI, benefit both the peers who are included in the procedure and

children with ASD in numerous ways. For example, PNIs can lead to increased initiations of social interactions, the use of a variety of communication devices, improvement in language skills, and less isolation on the playground (Kamps et al., 2015; Kasari et al., 2011; Mason et al., 2014; Thiemann & Goldstein, 2004).

Previous research has also emphasized the importance of peer-play in children's development (Wolfberg et al., 2012). Positive peer-play experiences are associated with different aspects of social development in children (O'Keeffe & McNally, 2020), such as social interaction skills (Veiga et al., 2016; Zosh et al., 2017), social understanding (Stagnitti & Cooper, 2009), and the development of the theory of mind (Qu et al., 2015). In contrast to typically developing children, who mostly do not require motivation or guidance in how to play with their peers, children with ASD are usually confronted with a variety of obstacles to peer-play (Wolfberg et al., 2012). The social play of children with ASD differs from their typically developing peers in various contexts, and they often do not benefit from inclusion in peer-play experiences (Wolfberg et al., 2012). Thus, it is important to support children with ASD to encourage their active participation in peer-play (Wolfberg et al., 2012). For this reason, a technique called Integrated Play Groups (IPG; Wolfberg & Schuler, 1993) was developed to provide children with ASD with relevant and sufficient support for peer-play (Wolfberg et al., 2012).

IPG procedures involve encouraging children with ASD to engage naturally in play with their peer-partners, which is further facilitated by their teachers (Lantz et al., 2004; Wolfberg, 2008; Wolfberg et al., 2012). Within the IPG model, children with ASD are novice players while typically developing peers are expert players (Wolfberg & Schuler, 1993). The play sessions, each consisting of a group of three to five children, are designed to provide a sense of familiarity, belonging and regularity among children with ASD and typically developing peers (Wolfberg 2008; Wolfberg et al., 2012; Yang et al., 2003). Therefore, the core practices of the IPG model include play initiations, scaffolding play, guiding the social-communication, and guiding play (Wolfberg 2008; Wolfberg et al., 2012). Numerous research findings suggest that the IPG was positively effective in increasing social interaction and communication (Yang et al., 2003; Zercher et al., 2001), social engagement and play (Lantz et al., 2004; Wolfberg & Schuler, 1993; Wolfberg et al., 2015; Zercher et al., 2001), and joint attention and language (Wolfberg & Schuler, 1993; Zercher et al., 2001) for children with ASD. However, the current PMI literature indicates that teaching children with ASD to engage in social interaction can be difficult during play (Dotson et al., 2010; Thiemann-Bourque et al., 2017).

Therefore, PMIs such as PNI can be enforced along with the IPG in order to support the social interaction and communication needs of such children (Prendeville et al., 2006; Wolfberg et al., 2012). While the IPG and PNI are evidence-based practices for children with ASD, there is a key difference between the two procedures (DiSalvo & Oswald, 2002; Hume et al., 2021). The IPG was designed to support development of appropriate-age social play and social communication in children with ASD through a playgroup in a supportive environment without the leading of an adult (DiSalvo & Oswald, 2002; Yang et al., 2003). The PNI as a component of PMIs was designed to build a social network of typically developing students along with their peers with ASD with adult direction to naturally facilitate social interactions

and social behaviours among them (DiSalvo & Oswald, 2002; Kamps et al., 2015). Accordingly, integrating IPG with PNI might have a positive impact on improving social behaviors for children with ASD.

This study expands on the existing research literature by incorporating PNI as a component of PMIs, and the use of IPG procedures to evaluate their effectiveness at improving basic social communication skills (i.e. initiations and responses) and positive social behavior (i.e. the ability to greet or to ask for something) in three children with ASD. More specifically, this study tries to answer the following research questions:

1. Does the implementation of a PNI with incorporated IPG lead to a longer duration of social interaction between children with ASD and their peer-partners?
2. Does the implementation of a PNI with incorporated IPG lead to an increase in positive initiations and responses between children with ASD and their peer-partners?

Method

Participants

Participants consisted of three nine-year-old children with ASD, nine typically developing peers, and three teacher-facilitators. The children with ASD were enrolled as second-grade students at an inclusive elementary school: Fox (male), Sam (male), and Nia (female). The typically developing peers were all eight years old at the time of the study. Two written informed consent forms were obtained. One was from the parents of each of the children participating in the study and another was from their teachers.

Children with ASD

The three children with ASD met the following criteria for study inclusion: (a) aged 8 to 9 years, (b) only diagnosed with ASD, (c) their Childhood Autism Rating Scale-2nd edition scores (CARS-2; Schopler et al., 2010) were mild to moderate (30–36.5). (d) IQ scores above 90 (WPPSI-IV; Wechsler 2012), (e) had a moderate to weak level of social communication and interaction skills based on teachers' observations and records, (f) limited peer interaction skills, and (g) enrolled in the second grade of an inclusive elementary school.

Fox was a 9-year-old Arab-American male diagnosed with ASD whose first language was Arabic based on his school file. His CARS-2 score was 31, his IQ score was 98, and he displayed low levels of academic and social functioning, and spent most of his school day in the general classroom. He performed well in his assignments such as math and writing, but experienced difficulties focusing on assignments and participating in classroom activities. For example, he obtained good results on his math and writing assignments. However, based on his teachers' and parents' reports, Fox struggled to interact with his peers and lacked appropriate social com-

munication skills, especially how to start a conversation, get involved in a game, how to maintain eye contact, and to imitate other children in the classroom. He spoke in one- or two-word utterances like "play here; play with toy". He participated in math, reading, writing, and play classes in the second-grade general classroom, but he took other subjects in the special education room for individual teaching.

Sam was a 9-year-old Arab-American male whose first language was Arabic at the time of the study and who was diagnosed with ASD based on his school file. His CARS-2 score was 30, his IQ score was 96, and he displayed low levels of academic and social functioning. He performed his assignments with some difficulty, such as math and writing, including having difficulties focusing on assignments and participating in other classroom activities. He struggled to interact with his peers and lacked appropriate social communication and interaction skills, especially difficulties starting a conversation, getting involved in a game, maintaining eye contact, and imitating other children. He spoke in one-to three-word utterances like "play; play here; play with toy". He participated academically in math, reading, writing, and play classes in the second-grade general classroom, the same as Fox. He took other subjects in the special education room for individual teaching.

Nia was a 9-year-old Arab-American female. She was diagnosed with ASD and her first language was Arabic based on her school file. Her CARS-2 score was 34, her IQ score was 98, and she displayed moderate levels of academic and social functioning. She also experienced difficulties participating in classroom activities, missed social initiations and responses within her class, and spent most of her school day in the general classroom. She lacked skills starting a conversation, getting involved in a game, maintaining eye contact, and imitating other children in the classroom. Nia spoke in two- or three-word utterances like "I want toy; I play." She participated academically in math, reading, writing, and play classes and took other subjects in the special education room for individual teaching.

Typically Developing Peers

The group of typically developing peers comprised nine second-grade classmates of Fox, Sam and Nia. There were three girls and six boys, all of whom were eight years old at the time of this study. According to the peer selection criteria, students were selected based on a general education teacher's recommendation that they had good academic performance, good social skills, and clearly spoken with understandable speech. Overall, the teachers identified children with excellent social and communication skills who were capable of leading group interactions during the intervention.

Teachers

One general education teacher and two special education teachers were included in this study, all of whom taught second-grade children at inclusive elementary school. The general education teacher had a bachelor's degree in elementary education, had taught reading and communication classes to second-grade students, and had six years of teaching experience (two of them spent teaching children with ASD in elementary inclusive classrooms). The two special education teachers had bachelor's

degrees in special education, with a concentration on teaching children with ASD, and had four years of teaching experience. As had the general education teacher, they had spent two years teaching children with ASD in elementary inclusive classrooms.

Setting

The children's regular classroom during free play served as the inclusive setting for this study. At certain times of the day, their classroom was divided into four areas of the same size: the library, the art space, the play area, and assistive technology. There was one classroom suited to this study, and each child with ASD was paired with a group of three typically developing peers at different times.

The library area contained a variety of children's books and stories. The teachers and children used the library for reading as well as other functional goals such as learning how to read out loud in front of their peers and teachers, providing opportunities for children with ASD to socially interact with their peers and encouraging them to imitate their peers' social and communication skills. Teachers used the art area to provide children with opportunities to reflect on their feelings or thoughts through drawing, to teach and reinforce positive social behavior, and to provide children with ASD with opportunities to participate and communicate with their peers in a natural setting. The art area provided children with ASD with an opportunity to draw, communicate, and tell others about their feelings. The play area aided the teachers in instruction, incorporating toys, games, or small-group activities in, for example, math lessons. Furthermore, teachers used the play area to reward or incentivize positive social behavior, such as allowing children to play with toys or watch a movie for a limited time if they completed their work early or correctly. The play area also provided opportunities for children with ASD to engage with their peers and participate in activities together in a natural situation, such as watching a movie. Finally, the assistive technology area contained computers and devices such as communication equipment for children with ASD to communicate with their classmates and teachers. These devices helped capture children's attention during lessons and provided opportunities for children to interact by playing educational video games together, for example. Class instruction, along with the teaching and practicing of communications skills and positive social behavior also took place there.

The classroom schedule included five classes per day of 45-minutes each. In the morning, children initially sat in the library area, listening to each other read or tell stories. Then, they attended a math class, followed by a language class. In the afternoon, after a 30-minute snack break in the playground, the children attended science class and then reading, the last class of the day. During that time, teachers provided one-on-one assistance with daily homework, and children played with video games or other games in the play area. There was a total of 12 students in the classroom. Only nine classmates of the three children with ASD participated in this study. While the play groups were occurring, other students were provided one-on-one assistance with daily homework by a teacher in the classroom.

Procedure

Experimental Design

A single-subject concurrent multiple baseline across participants was used to examine the effects of the PNI with an incorporated IPG model. Within this type of research design, the experimenter starts by measuring a trait of interest, then applies a treatment before measuring the same trait (Christ, 2007). In this study, the social skills of children with ASD were observed in three conditions: baseline, intervention, and maintenance conditions.

Measurement

The basic social communication skills and social behaviors of the children with ASD were recorded using Social Interaction Code (SIC; Nienmeyer & McEvoy 1989). The SIC is used to record initiations (motor or vocal behavior directed to a peer to evoke a response), responses (a reply within three seconds, which reflects reciprocity), and the length of time the children spend in interaction (through 0 to 300 s). It is a six-item social skill rating scale used to assess the presence of particular social behaviors including being quiet during play, being part of the group, interacting with others, making eye contact, listening to the speaker, and asking for something before taking it. The SIC was modified for this study, and utilized a three-point Likert scale where 0 indicated the behavior never appeared; 1 indicated the behavior seldom appeared; 2 indicated the behavior sometimes appeared, and 3 indicated the behavior always appeared.

The research design targeted positive social behaviors (i.e., positive initiations and positive responses).Positive initiations were defined as any verbal or motor behavior directed at a peer, such as asking to share materials or to join others in playtimes by saying “can”, “may I play”, or “please” or to request assistance from a peer by saying “please help me” or “please”.Positive responses were defined as any verbal or motor behavior resulting from an initiation, such as sharing, playing, or talking to another child, or listening to another child speaking (e.g., making eye contact), and/or accepting comments and requests from others by saying “ok”, “yes”, or nodding of the head. The methods of positive initiations and responses were calculated and graphed for each child during each baseline and intervention.

The SIC was utilized to collect data during the baseline and intervention conditions.During observations, researchers focused on social communication behaviors such as initiations and responses to peers, and positive social behaviors including verbal greetings such as “hi” or “hello” or non-verbal greetings such as hand waving. Basic social behaviors were recorded during the social behavior and social communication skills assessment, including greetings or requests, listening to another student when he or she is speaking (eye contact), and accepting peers’ comments and requests.

Baseline

To establish baseline data on the basic social communication skills and social behavior for Nia, Fox, and Sam, nine to 17 play sessions (20-minutes each) that resembled those that were part of the school curriculum were arranged for each of the three playgroups. These sessions were held at different times in the same classrooms.

As part of the regular play environment, teachers maintained the responsibilities they normally held during daily play sessions, including giving instructions to children when needed, teaching when "teachable moments" emerged during play, and monitoring the group to prevent negative social behavior. During the baseline play sessions, the teachers were instructed not to modify their teaching or other behaviors and they did not introduce incentives or activities to promote peer instruction or interaction between children. Given that this was an inclusive school where the curriculum and teaching methods were designed to be inclusive of children with ASD, these teachers already had training in social communication skills development for children with ASD.

A graduate research assistant and the investigator observed all baseline play sessions and collected observational data on interactions between children with ASD and their typically developing peers.

Teacher Training

Ethical approval to conduct the study was obtained from the university. Then, the principal of an elementary school in which students with ASD were educated in inclusive classrooms was contacted to select as study potential teachers of the students with ASD who would be participating in this study. Three teachers were candidates to participate. The investigator met with the three teacher-facilitators to explain the study goals and procedures as well as to provide informed-consent forms. Following completion of the forms, proper training was provided to the three teacher-facilitators.

To provide proper training, the investigator met with the three teacher-facilitators to explain the study goals and procedures as well as provide informed consent forms. This was a one-time, 100-minute training session during which the following were addressed: (a) how to train the typically developing children to work with their peers with ASD; (b) how to set and facilitate the natural scenes for play and interaction between peers and children with ASD and; (c) how to observe the interaction and offer support only as needed. Video clips were used to educate the teachers on how to train typically developing peers to implement the intervention. The training package consists of the peer-training program. The teachers were given strategies on setting up and facilitating natural scenes for play and interaction between typically developing peers and children with ASD. For example, teachers were shown how to arrange the play area and select toys suitable for interaction between peers and children with ASD. The arrangement of the play area should be as normal to meet the needs of children with ASD and to avoid any unexpected behaviors. Further, teachers were

trained on how to observe the interaction and to offer support only as needed while the children played. The teachers were trained only to observe the interaction and not to interfere unless inappropriate behaviors occurred.

Peer Training

The peer training was designed and the content selected based on several previous studies of PNIs (Harrell et al., 1997; Kamps et al., 1997, 2014, 2015; Kasari et al., 2006; Thiemann & Goldstein, 2004) and on IPGs (Wolfberg et al., 2012; Wolfberg & Schuler, 1993; Yang et al., 2003; Zercher et al., 2001).

The three teacher-facilitators used a researcher-designed training program with nine typically developing peers to teach them how to work with children with ASD to initiate interactions and practice positive social behavior. Initially, nine peers were separated into two groups (five peers in one group and four in another group). The peers training took place over two weeks at the school, with two 45-minute sessions per week. During all training sessions, time was set aside for peers to practice the skills were taught. Training content included a discussion about how children can be different, why that may happen, and an overview of challenges which children with ASD may experience during social interaction such as: interpreting facial expressions, sharing, playing, turn-taking, and social communication. The typically developing peers were taught different strategies by teachers-facilitators for facilitating successful interactions with children with ASD and the communication skills that they could teach their peers with ASD during the intervention.Further, they were taught how to use verbal prompting strategies in order to encourage their peers with ASD to initiate or respond positively.The typically developing peers also were trained on how to use positive verbal reinforcement when a child with ASD responded positively.

Pictures and video clips were used to train typically developing peers through direct instruction and video modeling. The pictures and video clips were used to explain how some children have different ways of speaking and communicating and to show the typically developing children how to use five discrete steps to greet, communicate, and interact with children with ASD. The five steps included: (1) sitting next to or across from the child with ASD; (2) attempting to get the attention of the child with ASD using verbal prompting by calling his or her name, waving a hand, etc.; (3) initiating a greeting with “Hello”; (4) waiting for a response such as waving a hand, making eye contact, etc.; (5) positively verbally reinforcing the child with ASD by saying “Good” if a response was given, (6) repeating steps 2–5. The typically developing peers were taught to keep repeating steps 2–4 if they got no response from their peer with ASD.

The pictures were utilized at the beginning of each training session. Then, the video clips were used to clearly show how to model the five discrete steps within a play scenario. The video clips showed actual children with ASD and those with communication difficulties, for example, showing how a child with ASD plays with their toys with other peers. In the last 15 min of each training session, typically developing peers were given free time for independent play to ensure they would be able to train their peers with ASD during playtime. During the final time, the typically developing peers were asked to practice the steps of the intervention and were given feedback

under observation from their teacher-facilitators. A checklist of the planned procedural steps was used as mastery criteria to ensure that the typically developing peers would be able to train their peers with ASD (See Appendix A).

Intervention

The intervention was implemented in an inclusive second-grade classroom using games and age-appropriate play activities through peer-led play. Altogether, one child with ASD, one teacher, and three peers were assigned to each network, rotating the teachers and peers throughout the ten sessions for each group to minimize group effects. The teachers and typically developing peers only were rotated throughout the sessions, while the students with ASD stayed in their groups. Not all the peers were shifted around, just one or two were, to avoid sudden behaviors might occur when children with ASD are confronted by changes in peers. The typically developing peers who were not sitting in close proximity to children with ASD were rotated in order not to break the social interaction had been built.

Firstly, the teachers-facilitators set scenes for interaction and play and offered strategic support including (a) arranging the play session, directing ,verbally/visually promoting the trained peers to begin or complete an ongoing interaction; and (b) rearranging an ongoing interaction if inappropriate behaviors occurred. The teachers-facilitators offered strategic support only as needed, took a passive role, and allowed the trained peers to lead the interaction. Then, within their peer networks, typically developing peers applied three PMI strategies: (a) initiations (typically developing peers initiated social interaction with their peers with ASD based on what they learned during the training (e.g. in joint play activities); (b) prompting and reinforcing, and (c) proximity – typically developing peers were seated close to a peer with ASD so the child with ASD could imitate the appropriate socially positive behaviors (Odom & Strain, 1984; Watkins et al., 2015). With strategic support from the teachers-facilitators, typically developing peers taught children with ASD several skills related to basic social communication, including (a) requests and shares, (b) comments about one's own play (activities, or personal actions with objects), (c) comments about others' play (activities, or peer actions with objects, (d) niceties (e.g., please, thank-you, compliments), and (e) play organizing (e.g., talk about ways to play, and taking turns).

Most of the major features that constitute the IPG model and other PNI models (e.g. Wolfberg et al., 2012; Kamps et al., 2014, 2015; Schmidt & Stichter, 2012) were maintained during this study.These included (a) an established, consistent schedule; (b) a natural, integrated setting; (c) a well-designed play space with a variety of age and developmentally appropriate toys and props for guided participation and full immersion in play; (d) a group including up to five children with one or two with ASD and; (e) playgroup facilitators who provided different levels of guidance. Over a two-week period, each of the three peer networks participated in ten play sessions, each 20 min long, for a total of 30 sessions (See Appendix B).

During the intervention, teachers-facilitators designed activities to promote interaction between children with ASD and their typically developing peers. Teachers facilitated play and allowed the typically developing peers to tutor the children with

ASD in basic social communication skills and positive social behaviors. These activities included games, character play, verbal and non-verbal greetings, toy play, a peer sitting next to the child with ASD, organizing all children together in one place, as well as providing children with rules for participating in a group, including how to play together and how to respect each other.

During play sessions, teachers instructed typically developing peers to initiate activities as they were taught during training (using a five-step process). For example, during an activity designed around toy play, teachers provided the appropriate props, including small cars and toy cubes, among others. The teacher sometimes prompted one of the three typically developing peers to initiate interaction with a child with ASD by saying "Hello" or "Let's play with a car" or another toy. The typically developing peer then waited for the child with ASD to respond in some way, such as eye contact, hand waving, head nodding, or picking up the toy. After the child with ASD responded, all the children began to play together.

A similar procedure was applied to character play. After prompting from the teacher, a typically developing peer initiated interaction with a child with ASD through character play by choosing a character to perform, saying, for example, "I am going to be Spiderman!" The peer then waited for the child with ASD to respond in some way, such as through eye contact, hand waving, head nodding, or engaging in character play. The other two typically developing peers then responded to the initiation by choosing their own characters to perform and began to play together. As facilitators of the playgroups, the teachers had a variety of responsibilities during the play sessions. They promoted natural interactions between peers and children with ASD and encouraged interactions by prompting when needed. They also monitored the interactions between children to prevent negative social behaviors and compulsive initiations, intervening to stop such behaviors if they emerged (Harper et al., 2008; Jacob et al., 2009; Sevin et al., 2015). Overall, the teachers emphasized observing rather than engaging with children to allow typically developing peers to direct interaction as much as possible, allowing the children to interact naturally without interference.

The teachers also had the responsibility of ensuring activities were free from compulsive initiations. Compulsive initiations include instances where the peer repeated the initiation or held the hand of child with ASD after that child had refused to respond. Compulsive initiations were avoided because they can be interpreted as coercive by a child with ASD who can feel pushed into doing something he or she does not want to do, resulting in distress (Jacob et al., 2009; Sevin et al., 2015). During the training sessions of typically developing peers, examples of compulsive initiations were discussed, and teachers-facilitators taught the typically developing peers how to avoid this behavior with children with ASD. Teachers were instructed to intervene only if typically developing peers engaged in compulsory initiations to prevent coercion of children with ASD and thus prevent triggering any distress.

Maintenance

The same baseline procedures were followed in the maintenance condition. This condition included four 20-minute-play sessions. These sessions were conducted once a

week for four-week periods two weeks after the last intervention session. During this condition, no directions were given to participants (typically developing children or those with ASD).

Reliability

Inter-observer Agreement (IOA). An inter-rater agreement measure was developed by comparing the data collected by the two observers: the investigator and the trained graduate research assistant. The two observers independently rated children during each session. To determine the inter-observer agreement, the sessions were videotaped. After observation of all sessions, the observers randomly selected and coded 20% of the videotaped sessions. The inter-rater agreement was computed by dividing the number of agreements by the number of agreements plus disagreements and then multiplying by one hundred to obtain a percentage. The IOA was calculated for each dependent variable (i.e., duration of social interactions, positive initiations, and positive responses). The data suggested that the agreement percentage ranged between 95 and 00% with an average of 98% for the duration of social interactions. The average IOA for the frequency of positive initiations was 96% and ranged between 90% and 00%. The average IOA for the frequency of positive responses was 95 and ranged between 90% and 100%.

Procedural integrity. Procedural integrity was evaluated by the investigator using a checklist of all planned procedural steps for all the videotaped baseline and intervention sessions (See Appendix A). The procedural integrity data was collected on teacher-facilitators and typically developing peers to ensure that the intervention was implemented with fidelity. The procedural integrity data were collected for teacher-facilitators and trained peers in combination because the intervention steps mostly were implemented by the trained peers while the teacher-facilitators were responsible only for designing activities to promote interaction between children with ASD and their peers and to prevent unexpected inappropriate behavior. The investigator marked the procedural integrity with a check mark (√) for each intervention step that was implemented accurately or an "X" if it was inaccurately done. The percentage of procedural integrity was calculated by dividing the number of steps that were implemented accurately by the total number of intervention steps and multiplying by 100. The percentage of procedural integrity was 97%.

Social Validity

Data for social validity of the intervention was collected from participating teachers and trained peers. A questionnaire was developed for the three participating teachers to measure the social validity of the intervention (Foster & Mash, 1999). It addressed teachers' perceptions of the intervention's effect on the children's communication skills and positive social behavior. It consisted of four statements in which teachers rated their perceptions of the intervention using a 5-point Likert-type scale where 1 indicated not at all feasible/not benefit was noted at all/never; 2 indicated rarely to use /slightly benefit was noted; 3 indicated not sure/undecided, and 4 indicated fea-

sible to use some of the time/moderately benefit was noted and 5 indicated feasible to use nearly all the time/benefited a lot/definitely.

Further, a focus group interview was conducted with the nine trained typically developing peers for measuring the social validity of the intervention. This interview consisted of short questions relating to the perceptions of the nine trained peers of their feelings about the acceptability of the intervention procedures on improving

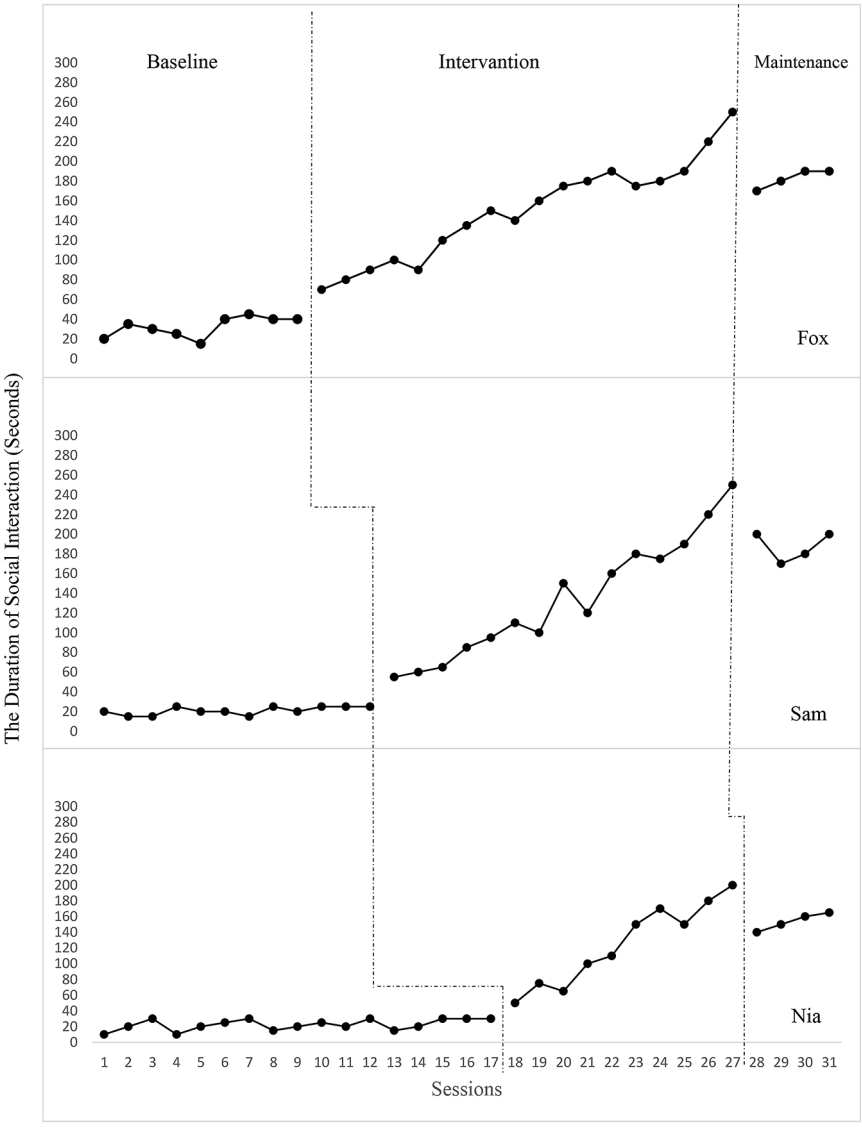


Fig. 1 The Duration of Social Interaction Across Three Children with ASD and Their Peers

the social behavior of their peers with ASD. The interview also was purposed to obtain feedback from the trained peers regarding the intervention and its effect on social behavior of their peers with ASD. The interview took approximately 30 min to complete.

Data Analysis

The effects of the PNI on the duration of social interaction, initiations of social interactions, and responses of children with ASD were investigated using visual analyses and by calculating effect sizes between baseline and the first intervention condition, and between the first intervention condition and the maintenance condition. The data points were analyzed by level, trend, and variability within conditions and across condition change for each child (Byun et al., 2017; Gliner et al., 2000). They were also analyzed for immediacy, overlap, and consistency of data between the baseline and the first intervention condition and between the first intervention condition and the maintenance condition (Horner et al., 2005). The Tau-U was used to calculate the intervention effect size for each child (Vannest & Ninci, 2015). The Tau-U is usually used in single-case studies, as it controls for positive baseline trends (Parker et al., 2011). To calculate the Tau-U, a web-based calculator for single-case research analysis was used (Vannest et al., 2016). As recommended by Parker et al., (2011), pre-intervention baseline Tau-U values above 0.40 or below 0.40 were deemed unstable and corrected, and this was repeated for all applicable baselines versus intervention contrasts. Condition contrasts were aggregated to provide an effect size for study participants, where: small effects: < 0.20, moderate effects: 0.20–0.60, large effects: 0.60–0.80, and large to very large effects: > 0.80 (Vannest & Ninci, 2015).

Table 1 Means, Standard Definitions and Ranges of Social Behaviors Across Participants

Dependent variable	Baseline			Intervention			Maintenance		
The duration of social interaction (in seconds)	M	SD	Range	M	SD	Range	M	SD	Range
Fox	32.2	10.35	15–40	149.7	50.57	70–250	182.5	9.58	170–190
Sam	28.7	7.73	15–40	133.3	48.43	55–200	170	15.00	160–180
Nia	22.3	7.10	10–30	125	52.18	50–200	153.75	11.08	140–165
Initiations									
Fox	3.60	0.89	2–4	8.20	1.16	6–10	7.20	0.50	7–8
Sam	3.00	0.96	1–4	7.20	1.15	5–9	6.50	1.30	5–8
Nia	2.80	0.73	2–4	7.20	1.32	5–9	6.70	0.50	6–7
Responses									
Fox	3.30	1.66	1–5	8.60	1.25	6–10	7.00	0.82	6–8
Sam	3.20	0.94	2–4	8.10	1.13	6–10	7.50	0.58	7–8
Nia	2.70	0.78	1–4	7.70	1.42	5–9	6.20	0.50	6–7

Results

Figure 1 depicts the duration of social interaction across conditions for the three children with ASD. Table 1 shows the mean scores, standard definitions and ranges of different types of social behavior across conditions by the targeted children with ASD. Overall results indicated a functional relation between the intervention and the duration of social interactions of children with ASD. An immediate positive change in level and an accelerating trend were noted for the duration of social interaction, with no overlap between baseline and intervention conditions for all three children.

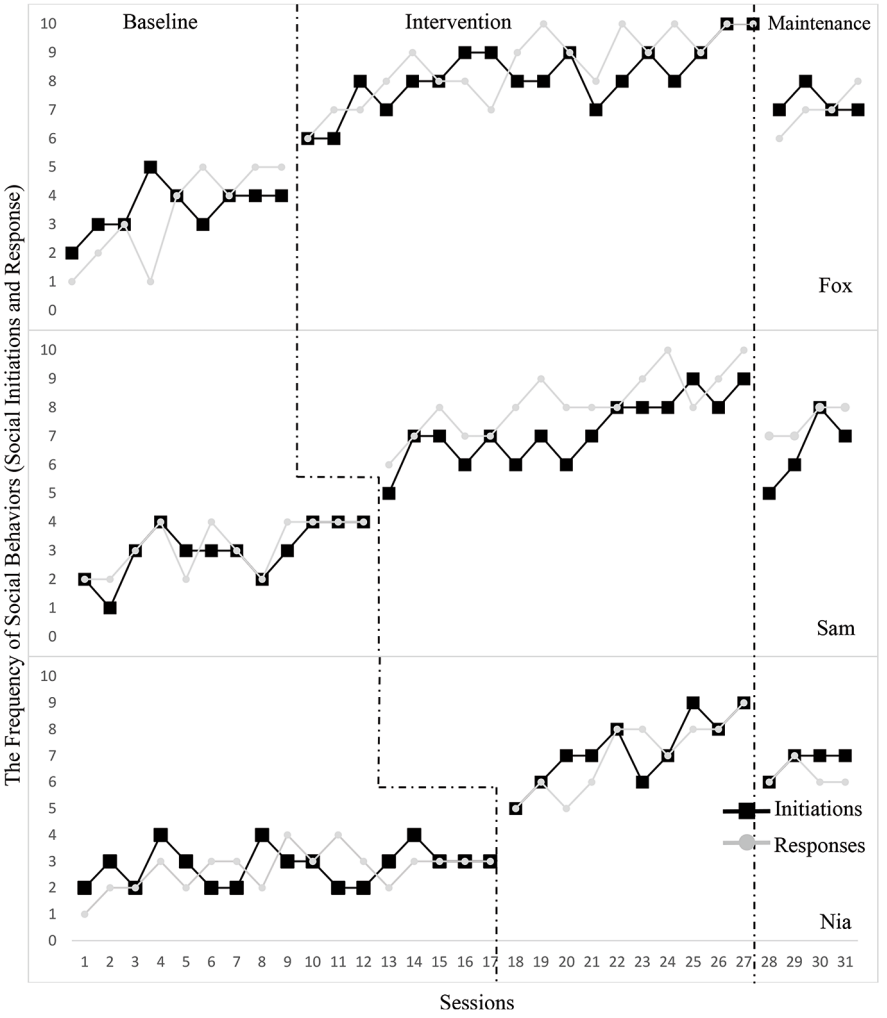


Fig. 2 The Frequency of Social Behaviors for Children with ASD

The Tau-U for the duration of social interaction was 0.90, indicating that the intervention had a large effect on the duration of social interaction.

Figure 2 shows the average frequency of social behaviors (initiations and responses) on a 6-item scale for different sessions. Overall results indicated that a functional relation between the intervention and the frequency of social behaviors was established with Fox and Nia. An immediate positive change was observed in both social initiations and responses and both remained stable during the intervention condition, with no overlap for all three children. The results showed a stable trend for all children during the baseline condition and a positive change in the intervention condition and in maintenance conditions. The omnibus Tau-U for the intervention was 0.91 for initiations and 0.90 for responses, indicating that the intervention had a large effect on both types of positive social behavior.

Fox. During the baseline, Fox engaged in the longest social interaction with peers (M = 32.2 s, range 15–40), with an unstable trend and with minimal variability. Meanwhile, an accelerating trend with high variability was observed during Fox's intervention condition. An immediate accelerating data trend between baseline and the intervention condition was demonstrated, and data indicated no overlapping between these conditions. Data also indicated a positive change from the baseline to the intervention condition(corrected baseline Tau = 0.91, $p = 0.00$), and a negative change from the intervention to the maintenance condition (corrected baseline Tau=-1.46, $p = 0.00$). The data levels were similar in the intervention and maintenance conditions and were consistent during the maintenance condition.

During the baseline, Fox engaged in positive social behavior and had the highest number of social initiations with peers (M = 3.6 s, range 2–5) and provided the highest number of social responses (M = 3.3 s, range 1–5), with a stable trend with minimal variability. A positive change was registered between the baseline and the intervention condition. A highly variable and unstable trend was observed during the intervention condition for both initiations and responses. Data showed no overlapping between the baseline and the intervention condition for both initiations and responses, suggesting immediacy of data. A positive change from the baseline to the intervention condition (corrected baseline Tau = 0.91, $p = 0.00$) was demonstrated for initiations, while from intervention to maintenance condition (corrected baseline Tau=-1.64, $p = 0.00$) a negative change occurred. Similar findings were obtained for social responses – a positive change from the baseline to intervention condition (corrected baseline Tau = 0.85, $p = 0.00$) and a negative change from the intervention condition to maintenance (corrected baseline Tau=-1.96, $p = 0.00$). Between the intervention and maintenance conditions, the data were similar and consistent, yet a decrease in initiations occurred during maintenance.

Sam. During the baseline, Sam engaged in shorter social interactions with his peers than did Fox (M = 28.7 s, range 15–40), and showed a stable zero-celerating trend with minimal variability. A positive change was noted between the baseline and PNI condition (corrected baseline Tau = 0.81, $p = 0.00$), and a negative level change from PNI to the maintenance condition (corrected baseline Tau=-1.23, $p = 0.00$). An accelerating trend, with higher variability, was noted during the intervention condition. The immediacy of data between baseline and the PNI condition occurred, the same

as with Fox, since the data indicated no overlapping between these conditions. Data levels were similar and consistent in the intervention and maintenance conditions.

During the baseline, Sam provided on average three initiations (range 1–4) and 3.2 social responses (range 2–4) with a stable zero-celerating trend and with minimal variability. A positive change in initiations was noted between the baseline and the intervention condition initiations (corrected baseline Tau = 0.91, $p = 0.00$) and a negative change between the intervention and the maintenance condition (corrected baseline Tau=-1.64, $p = 0.00$). Similar findings were obtained for responses with a positive change from the baseline to the intervention condition (corrected baseline Tau = 0.84, $p = 0.00$) and a negative change from the intervention condition to maintenance (corrected baseline Tau=-1.15, $p = 0.00$). An unstable trend with low variability was observed during Sam's intervention condition for both initiations and responses. Since the data did not overlap between the baseline and the intervention condition for both initiations and responses, the immediacy of data was observed. The level of data was similar between the intervention and the maintenance conditions and displayed consistency, yet there was a decrease in initiations during the maintenance condition, the same as with Fox.

Nia. During the baseline, Nia engaged in the shortest social interactions with her peers of all three children (M = 22.3 s, range 10–30), and showed a stable trend with minimal variability. A positive level change was noted between the baseline and the intervention condition (Tau = 1, $p = 0.00$), and an insignificant level change from the intervention to the maintenance condition (Tau = 0.25, $p = 0.48$). An unstable trend, with high variability, was noted in Nia during the intervention condition. The immediacy of data between the baseline and the intervention condition was displayed, the same as with the other two children since the data did not overlap between these conditions. Data levels were similar in the intervention and maintenance conditions and consistent in the maintenance condition.

During the baseline, Nia provided an average of2.8 initiations (range 2–4) and 2.7 social responses (range 1–4) showing a stable trend with minimal variability in both initiations and responses. A positive change was observed between the baseline and the intervention condition (Tau = 1, $p = 0.00$), and an insignificant change in initiations from the intervention to the maintenance condition (Tau=-0.23, $p = 0.52$). For responses, a similar finding was obtained -- a positive change between the baseline and the intervention condition (Tau = 1, $p = 0.00$), and an insignificant change from the intervention to the maintenance condition (Tau=-0.36, $p = 0.36$). An unstable trend with low variability was observed during Nia's intervention condition for both initiations and responses. The immediacy of data was observed due to no overlap between the baseline and the intervention conditions, for both initiations and responses. Data levels were similar in the intervention and the maintenance conditions, yet initiations declined during the maintenance condition and remained at consistent levels.

Social Validity

The three teachers completed the social validity questionnaire. Overall, their responses indicated that the intervention had improved the basic social communication skills of all three children with ASD. The teachers also expressed satisfaction

with the effectiveness of the intervention in the children's social behavior after the intervention. The satisfaction was high with a mean of 4.80. They also recommended using the intervention when teaching such students.

The focus group interview was conducted with the nine trained typically developing peers for measuring the social validity of the intervention.The findings of the interview indicated that the nine trained peers had positive acceptance of the intervention impact on the social behaviors of their peers with ASD.The feedback provided by the trained peers showed agreement about improving social interaction with their peers with ASD due to the intervention applied. They were also satisfied with the acceptability usage of the intervention procedures with their peers with ASD in their classrooms. The trained peers also felt comfortable playing with their peers with ASD after the intervention. They also expressed positive feelings about future usage of the intervention procedures with peers with ASD.

Discussion

Deficits in social and communication skills in children with ASD could interfere with their potential to benefit from a variety of social opportunities that are provided in the school setting, which includes playtime. One issue for the children with ASD is not just inclusion in different social activities with their peers, such as playtime, but also ensuring that specific interventions promote social interaction. Playtime could be challenging for children with ASD, due to limited adult supervision and structure. As can be seen, recognition of evidence-based practice which could be implemented in this setting is highly important, since this type of activity has numerous benefits for developing social-emotional skills (Wolfberg et al., 2012).

This study investigated the effects of PNI with integrated peer-play on communication and social skills in three children with ASD in an elementary school setting. Results demonstrated that the application of PNI increased the duration of social interaction among all three children with ASD and their peers. Results also indicated that the children's positive social behavior increased, specifically initiations and responses, from the baseline to the end of the intervention sessions, with large effect sizes in all three children. Although their positive social behavior (initiations, responses, and the duration of social interaction) slightly decreased during the maintenance condition, all three children showed an improvement compared to the baseline. Hence, this study demonstrates that the PNI with incorporated IPG has stable effects after the intervention. Participating teacher's high satisfaction with the intervention provides additional insight into its potential usefulness in inclusive classrooms.

The level and mean of social interactions of children with ASD increased substantially from the baseline to the intervention conditions. All target children spent more time interacting with their peers in the intervention and maintenance conditions compared to baseline, although they showed a slight decrease in social interaction during the maintenance condition. Moreover, all three children with ASD used words more frequently to interact with their peers during the intervention and maintenance conditions. For example, during the baseline, Fox would only share his toys or other possessions with his peers when prompted by his teachers. During the intervention,

he was able to ask his peers if they would like to borrow his toys during playtime. In addition, he could say 'hey', 'play' or point to a toy and wave his hand to socially initiate with his peers. Before the intervention, Fox would only play alone but after the intervention, he was able to share his toys, engage in small-talk, and play with one of his peer-partners. For example, he would make eye contact and listen to peers as a response to peer-initiation.

Before the intervention, Sam was unable to ask for help; within the intervention, he would say 'please help me' when he needed help during playtime. At the baseline, he was unable to make eye contact or ask to join others in playtimes; however, during the intervention, he made eye contact and asked to join and share with others during playtimes by initiating with his peers. Sam lacked appropriate social communication and interaction skills before the intervention. He had difficulties responding to peer-initiation such as starting a conversation, getting involved in a game, and making eye contact with other children. After the intervention, he could respond socially to peer initiations by listening to his peers, starting a small conversation (e.g. 'yes', 'ok' 'let's play'), or nodding his head or hand movements and making eye contact while sharing toys or playing with peers.

Before the intervention Nia also experienced difficulties socially initiating with her peers, such as asking to share materials or getting involved in a game. After the intervention, she could socially initiate with peers by making eye contact to get others' attention, requesting help, sharing, or joining in play by saying 'please help me' or 'please, I want toy or play'. Prior to the intervention, Nia would consistently avoid looking at her peers when they were speaking; during the intervention condition, she generally looked at her peers when they were talking and showed that she was paying attention to them. Nia was also able to say 'hello', 'yes, play', point to a toy, or nod her head or wave her hand to answer peer initiations. The social validity measure affirmed these findings, as the results of the teacher questionnaires indicated that all three teachers agreed on the improvement they observed in the basic social communication skills of the three children with ASD after the intervention.

These findings are similar to those obtained by Owen-DeSchryver et al., (2008), Banda & Hart (2010), Mason et al., (2014), McFadden et al., (2014), and Lee & Lee (2015) and demonstrated that the intervention had a positive effect on social and communication skills in children with ASD. This study confirms the benefits of including PNI to support children with ASD, using social skills in a natural setting via prompting, reinforcing, and modeling Kamps et al., 1997; Koegel et al., 2012a, b; McFadden et al., 2014). Although a greater number of peers could lead to greater variability of modeled social skills, this variability is constantly present in the social setting of the school population. This variability in peers might cause a greater generalization of newly learned or improved social skills (Gresham, 1998) than would a situation where only one peer is interacting with a student with ASD (Pierce & Schreibman, 1995; 1997).

This study demonstrated that IPG integrated with PNI enhanced the initiations and responses of children with ASD with their peers. Strain & Odom (1986) suggested that increased responses in target children could be a product of social reciprocity since positive peer engagement might lead to a positive response from the target child.

Moreover, it is possible that increased peer initiations, integrated play, and scheduled daily interventions helped children with ASD become more efficient initiators and responders to social interaction, since they had numerous opportunities to practice their social skills (Owen-DeSchryver et al., 2008). Children with ASD were not trained to initiate interaction with their peers prior to the study, yet their initiations increased due to the intervention. One plausible explanation is that different environmental features (i.e., selection of activities, the natural environment of the classroom) provided the structure that children with ASD require.These features further provided an opportunity for them to engage in more social interaction. This is one contribution of the present study which used PNI and IPG with elementary school children with ASD in the natural setting of the children's own school and regular classrooms where different aspects of IPG were applied. This finding would support implementing natural agents of intervention to effectively obtain improvement in the social behaviors of children with ASD. Previous research suggested that applying effective interventions in natural settings with typically developing peers yields the most powerful improvement of the core deficits of children with ASD (e.g. social, communication and interpersonal skills; Kamps et al., 2002).

Limitations and Recommendations for Future Research

This study had several limitations. First, while single-case design studies, particularly on children with ASD, usually involve small samples, this study has particularly limited generalizability, since only three children were involved. Future research should replicate this type of intervention in children with ASD by varying types of settings and by including children with different levels of functioning. Second, while this study included only peers who were trained on how to interact with their peers with ASD, it could influence the use of naturalistic agents of intervention. Thus, future research might assess the generalizability of social behavior when trained and untrained peers are engaged. Future research could also collect data about interactions that typically developed peers engage in during the day, to control for potential bias in their interactions with their peers with ASD. Third, given the importance of demonstrating lasting effects of PMIs (Koegel et al., 2012a, b; Laushey & Heflin, 2000; Thiemann & Goldstein, 2004; DiSalvo & Oswald, 2002), it would have been beneficial in this study to determine whether the learned skills remained intact by testing the children with ASD over time.Fourth, the visual analysis indicated that one participant had an ascending baseline, suggesting that the functional relationship between the intervention and improved communication skills is not easily determined. Fifth, the use of the Tau-U statistic to determine effects sizes has several important limitations (Brossart et al., 2018), including inconsistent Tau-U terminology in published single-case studies, math problems that cause unexpected results (e.g. Tau-U exceeding −1 and +1), difficulties with visual representation, and weak correlation between Tau-U effect size statistics and visual analysis. Finally, social validity data of the intervention was obtained only from participating teachers and trained peers. Therefore, future research should incorporate gathering social validity data from participants with ASD in order to establish the significance of the intervention from direct recipients.

Conclusions

This study supported previous literature by demonstrating that PNIs directed toward social and communication skills can be effectively adopted during routine activities in an inclusive classroom within a natural school setting. It also extends previous research by adopting an IPG model and showing that nine-year-old children with ASD are capable of learning the target skills required for social interaction with their peers during play and with minimal adult facilitation. Since the children with ASD had poor social and communication skills prior to the intervention, the current study emphasizes the need for incorporating effective interventions in elementary schools so children with ASD can gain socially from an inclusive educational placement.

Further, the findings of this study highlight the importance of natural agents of interventions in the development of functional communication and social behaviors in children with ASD. The findings also support the evidence that using the natural agents of interventions from a child's environment can effectively develop communication and social skills (Hume et al., 2021). Employing natural factors from the environment of children with ASD, including their typically developing peers, play routines, and toys in teaching children with ASD would increase the children's learning. This would also help children with ASD to simultaneously learn and practice in natural environments and situations. Therefore, teachers of children with ASD should consider implementing natural agents of intervention in their daily teaching practices.

Appendix A

The Procedural Fidelity Checklist.

Step	Mark (√) if observed or (X) if not observed.
1.The teachers-facilitator designed activities to promote interaction between children with ASD and their typically developing peers (organizing all children together in one place).	
2.The trained peer sitting next to or across from the child with ASD.	
3.The trained peer attempting to get the attention of the child with ASD using verbal prompting by calling his or her name, waving a hand, etc.	
4.The trained peer initiating a greeting with "Hello".	
5.The trained peer waiting for a response such as waving a hand, making eye contact, etc. from the child with ASD.	
6.The trained peer positively verbally reinforcing the child with ASD by saying "Good" if a response was given.	
7.The trained peer repeating steps 2–5 if no response from the child with ASD given.	

Appendix B

The Used Strategies of Peer Network Interventions (PNI) and Integrated Play Groups (IPG).

The Strategy	Model	Deliver
A group including up to five children with one or two peers with ASD.	PNI	Teacher
A well-designed play space with a variety of age and developmentally appropriate toys and props for guided participation and full immersion in play.	IPG	Teacher
Adult facilitation and repeated social learning opportunities in a natural, integrated setting with typically developing peers.	PNI	Teacher
Play initiations and scaffolding play.	IPG	Trained peer
1. Encouraging children with ASD to engage naturally in a playgroup.	IPG	Trained peer
2. playgroup facilitators who provided different levels of guidance.	IPG	Trained peer
3. Build a social network along with their peers with ASD.	PNI	Trained peer

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Declarations

Conflict of interest The author declares that there is no conflict of interests to disclose.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed Consent Parental informed consent was obtained from each participant included in the study.

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